Self Learning Assignment - 1

Topic : Comparison of Agile vs DevOps, Case study of JIRA and Asana in real projects, Writing effective user stories and acceptance criteria, Advanced requirement elicitation techniques (interviews, ethnography), Requirement traceability matrix (RTM), Tools for requirement management (e.g., IBM DOORS)

1. Introduction :

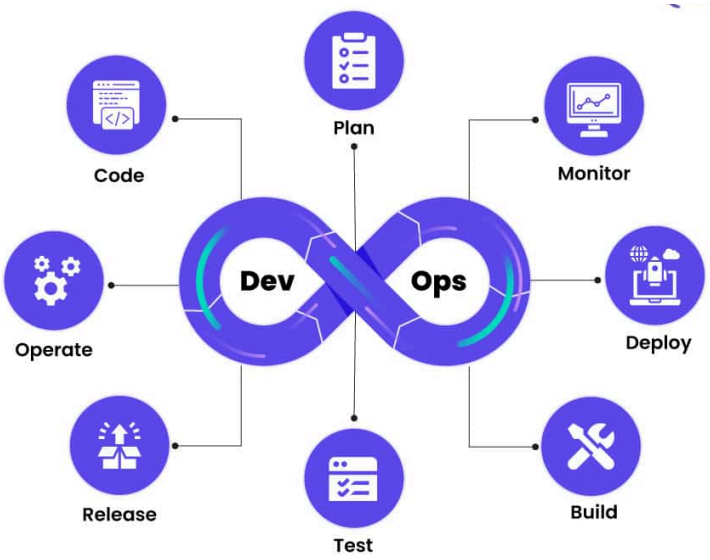
This self-study covers Agile and DevOps, comparison of Agile and DevOps, Jira and Asana in real project contexts, effective user stories and acceptance criteria, advanced requirement elicitation techniques, Requirement Traceability Matrix (RTM), and requirement-management tools such as IBM DOORS.

2. DevOps :

According to AWS, DevOps combines cultural philosophies, practices, and tools to increase an organization's ability to deliver applications and services at high velocity. It brings development and operations closer across the application lifecycle and emphasizes automation, collaboration, reliability, security and continuous delivery.

2.1 Key DevOps Practices :

1. Continuous Integration (CI): developers regularly merge code into a central repository and automated builds/tests validate changes.
2. Continuous Delivery (CD): code changes are automatically built, tested and prepared for release.
3. Infrastructure as Code (IaC): infrastructure is provisioned and managed using code and software-development practices.
4. Monitoring and logging: teams observe application and infrastructure behaviour to detect and address issues.
5. Microservices: applications can be organized as smaller, independently deployable services.

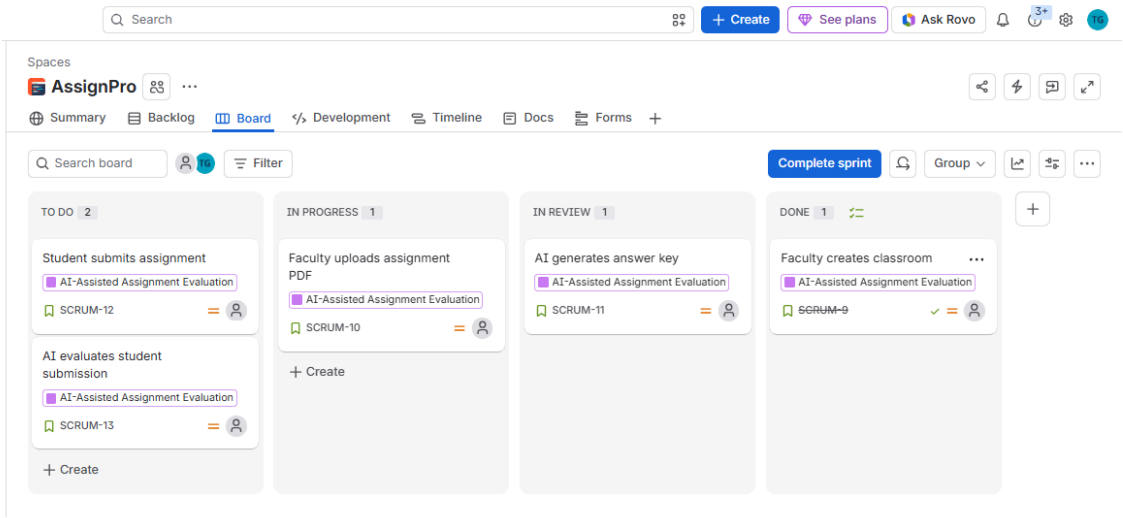


3. JIRA

Jira is an Agile project and work-management tool from Atlassian. It supports Scrum, Kanban, backlogs, boards, timelines, reporting and workflow tracking. Teams can use it to plan work, prioritize backlogs, track execution and report progress.

3.1 Important Jira Features :

- Product and sprint backlogs
- User stories, tasks and issues
- Scrum and Kanban boards
- Sprint planning and tracking
- Dashboards and reports
- Workflow/status tracking
- Integrations with development and collaboration tools



4. Agile vs DevOps :

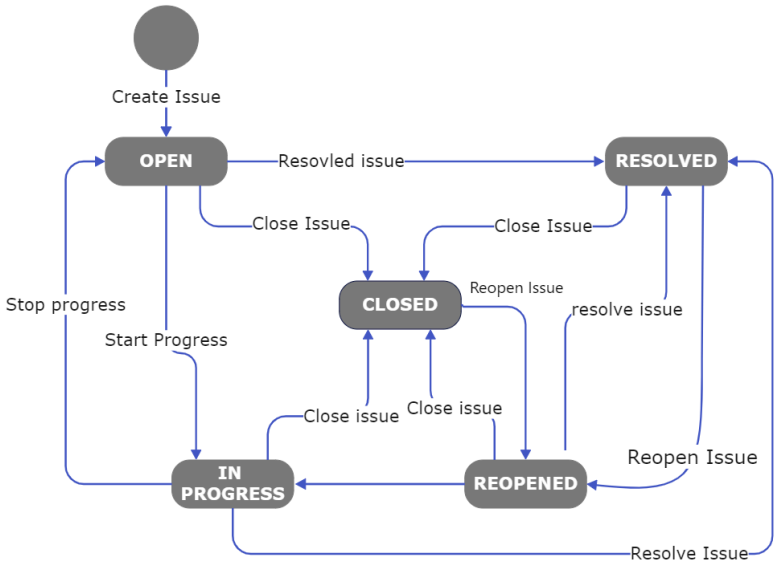
Agile and DevOps are complementary approaches. Agile primarily focuses on iterative product development, collaboration and customer feedback. DevOps extends collaboration across development and operations and emphasizes automation, continuous integration, delivery, deployment and operational feedback.

Aspect	Agile	DevOps
Primary focus	Iterative product development and customer value	Fast, reliable software delivery and operations
Teams	Product/development teams and stakeholders	Development, operations, QA, security and related teams
Delivery	Incremental delivery through iterations/sprints	Continuous integration, delivery and deployment
Key practices	User stories, backlog, sprint, review, retrospective	CI/CD, automation, IaC, monitoring, logging
Feedback	Customer/stakeholder feedback	Customer plus production/operational feedback

5. Case Study of Jira in a Real Project Context

Jira is designed to support teams that plan work in backlogs, select work for sprints, track issues through workflows and use reports to understand progress. A typical software team can create epics for large capabilities, break them into user stories and tasks, prioritize the backlog, plan a sprint, track work on a Scrum board and review sprint metrics.

Illustrative example : An online education team can create an Epic such as “Assignment Evaluation”, break it into stories such as “Faculty uploads assignment PDF” and “Student submits assignment”, prioritize the backlog, select stories for a sprint and track them from To Do to In Progress to Done.



6. Case Study of Asana in a Real Project :

Asana is a work-management platform used to coordinate projects and cross-functional work. In Asana's published Zoom case study, Zoom initially rolled out Asana to five departments. As adoption increased, teams used centralized project links and moved monthly business reviews into Asana. The case study also describes training and using Asana to teach repeatable project-management practices.

7. Writing Effective User Stories and Acceptance Criteria :

An effective user story expresses who needs a capability, what they need, and why they need it.
Format: As a [user], I want [feature], so that [benefit].

7.1 Characteristics of a Good User Story :

1. Clear and understandable
2. Focused on user value
3. Small enough for an iteration
4. Negotiable rather than an overly detailed technical specification
5. Testable through acceptance criteria

7.2 Example :

As a faculty member, I want to upload an assignment PDF so that the system can process it for AI-assisted evaluation.

7.3 Acceptance Criteria :

- Faculty can select a PDF file.
- The system validates that the file is a PDF.
- Unsupported file types are rejected with a clear message.
- The uploaded file is associated with the correct assignment.
- The uploaded document is available for the next processing stage.

8. Advanced Requirement Elicitation Techniques :

Requirement elicitation is the process of discovering, understanding and documenting stakeholder needs. Advanced techniques help uncover needs that may not be revealed by simple questionnaires.

8.1 Interviews :

Interviews involve structured, semi-structured or open-ended conversations with stakeholders. Questions explore goals, problems, current workflows, exceptions and desired outcomes.

Good practice : identify stakeholders, prepare questions, ask follow-ups, avoid leading questions, confirm understanding and document findings.

8.2 Ethnography :

Ethnography involves observing users in their real working environment to understand how tasks are actually performed. It can reveal informal processes, workarounds, environmental constraints and needs that users may not mention during interviews.
Example: observing faculty while evaluating assignments can reveal repetitive manual steps and decision points that may be missed in an interview.

8.3 Other Useful Techniques :

1. Contextual inquiry – interview combined with observation in the user's context.
2. Workshops – stakeholders collaboratively identify and prioritize requirements.
3. Prototyping – early interface/workflow models are used for feedback.
4. Document analysis – existing forms, policies and reports are examined.
5. Focus groups – representative users discuss needs together.

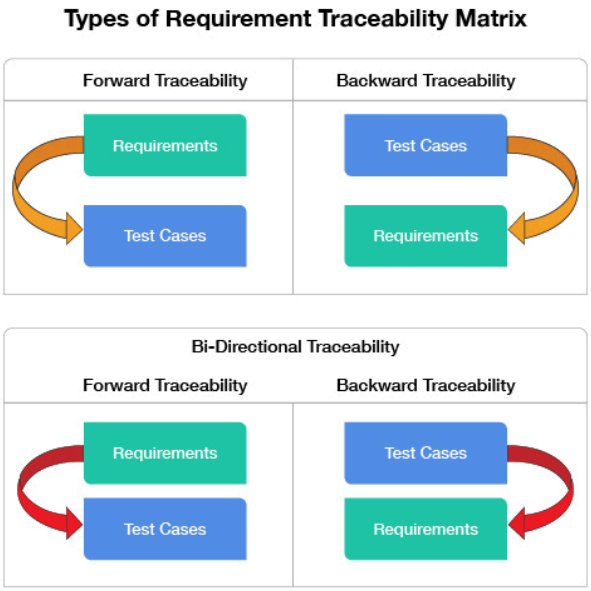
9. Requirement Traceability Matrix (RTM) :

An RTM maps requirements to related design elements, implementation items and tests. It helps verify that requirements are addressed and tested and supports impact analysis when requirements change.

Req. ID	Requirement	Module	Test Case	Status
REQ-01	Faculty can upload assignment PDF	Assignment Upload	TC-01	Implemented
REQ-02	System validates PDF type	File Validation	TC-02	Implemented
REQ-03	Student can submit assignment	Submission	TC-03	Implemented
REQ-04	AI generates draft evaluation	AI Evaluation	TC-04	In Progress

9.1 Benefits of RTM :

1. Ensures requirements are not missed
2. Links requirements to verification/testing
3. Supports change-impact analysis
4. Improves project visibility and documentation
5. Provides evidence of requirement coverage



10. Tools for Requirement Management :

10.1 IBM DOORS / DOORS Next :

IBM DOORS Next is a requirements-management solution used to capture, manage, trace and analyze requirements. It supports structured requirement information and traceability across the development lifecycle.

10.2 Common Capabilities :

1. Centralized requirement repository
2. Versioning and change management
3. Traceability between requirements and related artifacts
4. Review and collaboration
5. Baselines and history
6. Impact analysis and reporting

11. Tool Comparison :

Tool	Primary Purpose	Useful For	Key Strength
Jira	Agile project/work tracking	Backlogs, sprints, issues, Scrum/Kanban	Agile planning and development workflow
Asana	Work/project management	Cross-functional projects and coordination	Flexible project and task coordination
IBM DOORS Next	Requirements management	Requirements, traceability and change management	Structured requirements and traceability

12. Key Takeaways :

1. Agile focuses on iterative development and continuous stakeholder feedback.
2. DevOps connects development and operations through collaboration, automation and continuous delivery.
3. Jira is suited to Agile planning, backlog management, sprint tracking and reporting.
4. Asana supports broad work management and cross-functional coordination.
5. Good user stories should be clear, user-focused and testable through acceptance criteria.
6. Interviews and ethnography reveal stated and observed requirements.
7. An RTM provides traceability between requirements and downstream artifacts such as design and tests.
8. IBM DOORS Next supports structured requirements management and traceability.

13. References :

[1] Amazon Web Services, “What is DevOps?,” - https://aws.amazon.com/devops/what-is-devops/.

[2] Atlassian, “Agile Project Management,” - https://www.atlassian.com/jira/solutions/agile-project-management.

[3] Atlassian, “Jira Software,” Atlassian. [Online]. Available: https://www.atlassian.com/software/jira.

[4] Atlassian, “Agile Project Management,” - https://www.atlassian.com/agile/project-management.

[5] Asana, “Zoom: How Zoom uses Asana to manage work at scale,” - https://asana.com/case-study/zoom.